MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N6683, 1N6684, 1N6685, 1N6683US, 1N6684US, AND 1N6685US
JANTX, JANTXV, JANS, JANCH, AND JANCK

1. SCOPE

1.1 Scope. This specification covers the detail requirements for switching diodes, suitable for high stress environments where silver button construction may be inadequate (see 6.3). Three levels of product assurance are provided for each device type and two levels for chips as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-204-AH), figure 2 (D-5D), and figure 3.

1.3 Maximum ratings. T_A = +25°C.

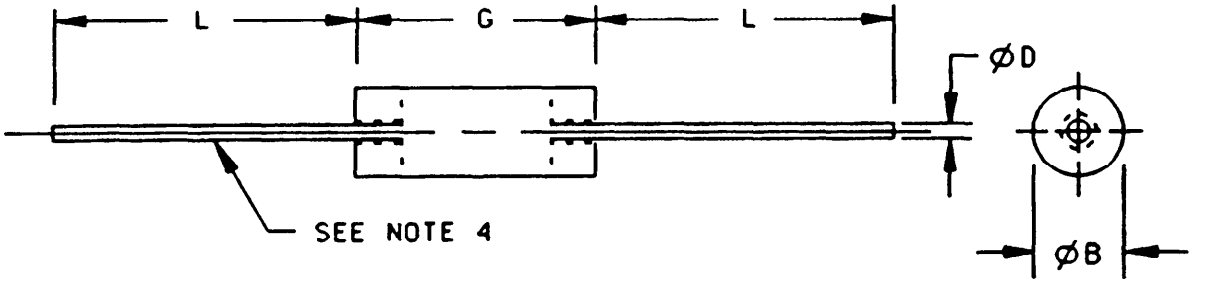
Types	V_{BR}	V_{RWM}	I_{O1} 1/ 2/	I_{FSM} tp = 1 μsec	T_j and T_{STG}	$R_{\theta JX}$ L = .375	$R_{\theta JEC}$ L = 0	$Z_{\theta JX}$
	V(pk)	V(pk)	mA	A(pk)	°C	°C/W	°C/W	°C/W
1N6683, 1N6683US	200	175	200	3.0	-65 to +175	160	50	25
1N6684, 1N6684US	250	200	200	3.0	-65 to +175	160	50	25
1N6685, 1N6685US	350	275	200	3.0	-65 to +175	160	50	25

1/ Derate at 3.0 mA/°C above T_L = 75°C at L = .375 inch.
2/ Derate at 4.0 mA/°C above T_{EC} = 125°C for US suffix devices.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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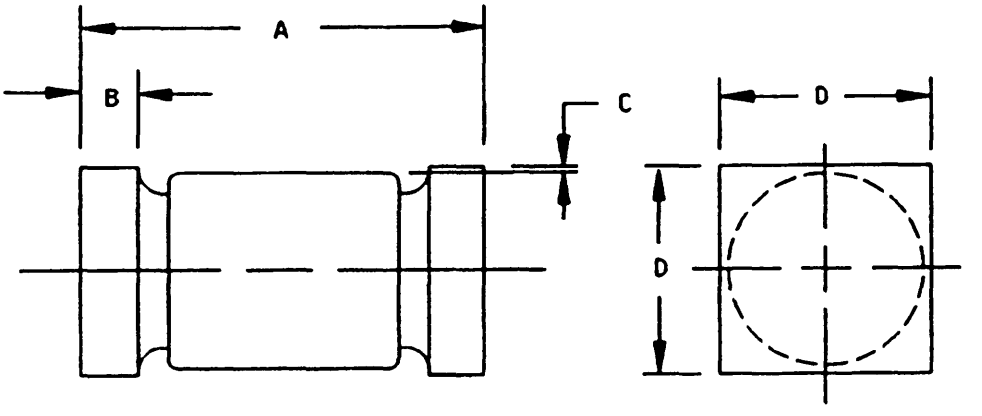
Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	1.52	2.29	.060	.090	2, 3
ϕD	0.46	0.56	.018	.022	4
G	3.05	5.08	.120	.200	
L	25.40	38.10	1.000	1.500	

TYPES 1N6683, 1N6684, AND 1N6685

NOTES:
1. Inch equivalents are given for general information only.
2. The maximum dimension of ϕB shall apply for dimension G.
3. The minimum dimension of ϕB shall apply over at least 1.65 mm (.065 inch) of dimension G.
4. The specified lead diameter applies in the zone between 1.27 mm (.050 inch) from the diode body to the end of the lead. Outside of this zone lead shall not exceed ϕB.
5. Dimensions are in millimeters.

FIGURE 1. Physical Dimension (DO-204-AH).

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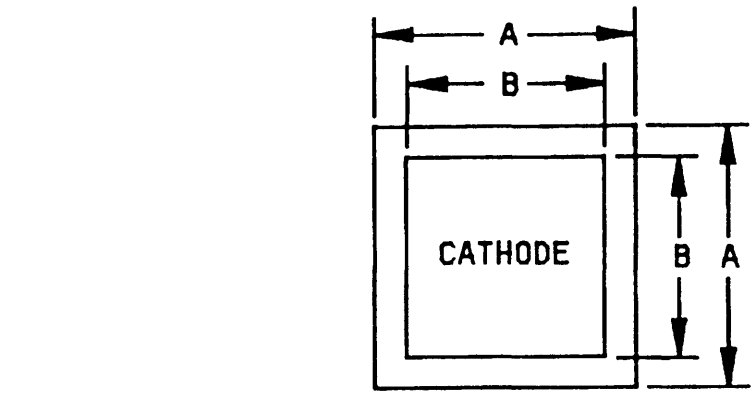


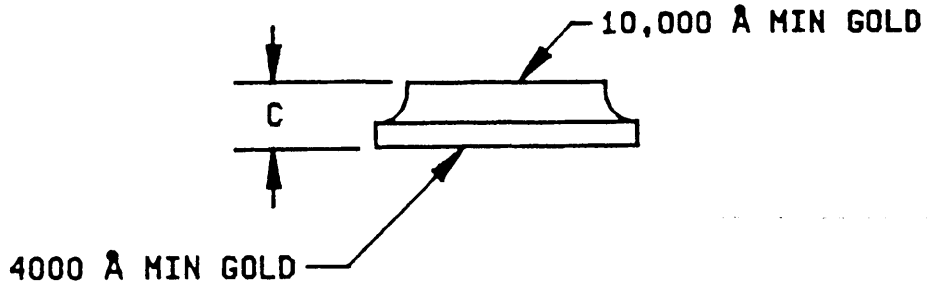
Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	4.19	4.95	.165	.195
B	0.48	0.71	.019	.028
C	0.08	----	.003	----
D	1.78	2.16	.070	.085

TYPES 1N6683US, 1N6684US, AND 1N6685US

FIGURE 2. Physical dimension for surface mount family (D-5D).

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Symbol	Dimension			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.019	.025	0.48	0.64
B	.008	.012	0.20	0.30
C	.007	.011	0.18	0.28

DESIGN DATA

Metallization:

Top: (Cathode) . . . . . Au
Back (Anode). . . . . Au

Au thickness:
Top: . . . . . 10,000Å minimum
Back: . . . . . 4,000Å minimum

FIGURE 3. JANCH and JANCK die (A Version).

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Types	V_{FM} at I_{FM} pulsed	I_{R1} at T_A = 25°C		I_{R2} at T_A = 150°C		t_{rr} $I_F = I_R$ = 30 mA I_{rr} = 3 mA	C_T V_R = 0
		nA dc	V_R	µA dc	V_R	nsec	Pf
1N6683, 1N6683US	1.0 V at 200 mA	100	175 V	100	175 V	50	5.0
1N6684, 1N6684US	1.0 V at 300 mA	100	200 V	100	200 V	50	8.0
1N6685, 1N6685US	1.1 V at 300 mA	200	275 V	200	275 V	50	8.0

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows.

T_{EC} . . . . . Temperature of end caps (mounting pads for US suffix devices).

EC . . . . . End cap.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, and 3 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.3.2 Diode construction.

a. Except for JANCH and JANCK, all devices shall be of metallurgically bonded, thermally matched, noncavity-double plug construction. Metallurgical bonds shall be in accordance with the requirements of category I (see MIL-S-19500).

b. The US version shall be considered structurally identical to the non-US versions except for lead attach.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking for US devices. For US version devices only, all marking (except for 3.5) may be omitted from the body, but shall be retained on the initial container.

3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US suffix devices, a minimum of three contrasting color dots spaced around the cathode end of the device may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV Levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c 1/	Thermal impedance (see 4.5.5)	Thermal impedance (see 4.5.5)
9	I_{R1}, V_{F1}, or V_{F2}	Not Applicable
11	I_{R1}, V_{F1}, or V_{F2} , $\Delta I_{R1} \leq \pm 100$ percent of initial value or ± 50 nA, whichever is greater. $\Delta V_{f1} \leq \pm 0.03$ V dc	I_{R1}, V_{F1}, or V_{F2}
12	See 4.3.1	See 4.3.1
13 2/	Subgroups 2 and 3 of table I herein; $\Delta I_{R1} \leq \pm 100$ percent of initial value or 50 nA, whichever is greater. $\Delta V_{F1} \leq 0.03$ V dc. Reverse scope display (see 4.5.2).	Subgroup 2 of table I herein; $\Delta I_{R1} \leq \pm 100$ percent of initial or 50 nA dc, whichever is greater. $\Delta V_{F1} \leq \pm 0.03$ V dc. Reverse scope display (see 4.5.2).

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this test.

2/ When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.1).

Method 1038, condition B, V_R = rated V_{RWM}; f = 50-60 Hz; I_O = 200 mA. An alternative of I_F (dc) = 200 mA may be used. T_A = room ambient as defined in the general requirements of MIL-STD-750, (see 4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.2.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
4	1037	V(pk) = rated V_{RWM}; I_O shall be a maximum of 200 mA, 2,000 cycles. T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); f = 50-60 Hz (see 4.5.1); ON time, OFF time, and I_O shall be adjusted to achieve a delta lead temperature at L = 3/8" from the body of 85°C, -5°C +15°C. Cycle time shall be a minimum of three minutes.
5	1027	I_O = 200 mA dc; V_R = rated V_{RWM} (see 1.3); f = 50-60 Hz (see 4.5.1); T_A = 150°C minimum. Increase T_A or I_O as required to achieve junction temperature as specified in MIL-S-19500.

4.2.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV or MIL-S-19500).

Subgroup	Method	Conditions
3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); V(pk) = rated V_{RWM}; f = 50-60 Hz (see 4.5.1); I_O = 200 mA. Note: Leaded samples from the same lot may be used in lieu of US suffix sample for life test.
5	3101 or 4081	See 4.5.3.
6	1032	T_A = +175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

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4.4.3.1 Group C Inspection, table V of MIL-S-19500

Subgroup	Method	Conditions
2	2036	Tension: Test condition A; weight 4 pounds, t = 15 seconds Lead fatigue: Test condition E. Note: Not applicable for US suffix types.
6	1026	T_A = room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); V(pk) = rated V_{RWM}; f = 50-60 Hz (see 4.5.1); I_O = 200 mA. Note: Leaded samples from the same lot may be used in lieu of U suffix samples for life tests.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Screening and life test. AC tests shall be conducted with a half-sine waveform of the specific peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.2 Scope display. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 20 μA per division. Reverse current over the knee shall be at least 100 μA and less than 200 μA. Any discontinuity or dynamic instability of the trace shall be cause of rejection of that device.

4.5.3 Thermal resistance. Thermal resistance measurement shall be performed in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the devices during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL}$ (max) = 160°C/W and $R_{\theta JEC}$ (max) = 100°C/W. The following conditions shall be applied if method 3101 is used.

I_H . . . . . 75 mA to 300 mA.

t_H . . . . . 25 seconds minimum.

I_M . . . . . 1 mA to 10 mA.

t_{MD} . . . . . 100 μs maximum.

LS = Lead spacing = 9.53 mm (.375 inch).

LS = 0 for US versions.

4.5.4 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.5 Thermal impedance ($Z_{\theta JX}$) measurements. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall not exceed the group A, subgroup 2 limit. The conditions for $Z_{\theta JX}$ testing are as follows:

I_H . . . . . 0.5 to 2 amps.

t_H . . . . . 10 ms.

I_M . . . . . 1 mA to 10 mA.

t_{MD} . . . . . 100 μsec max.

4.5.5.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Limits			
	Method	Conditions	Symbol	Min	Max	Unit
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.5	$Z_{\theta JX}$		25	°C/W
Forward voltage (1N6683 only)	4011	tp = 300 µS I_F = 200 mA	V_{F1}		1.00	V dc
Forward voltage (1N6684 only)	4011	t_p = 300 µS I_F = 300 mA dc pulsed	V_{F2}		1.00	V dc
Forward voltage (1N6685 only)	4011	t_p = 300 µS I_F = 300 mA dc pulsed	V_{F3}		1.00	V dc
Breakdown voltage	4021	I_R = 100 µA dc	V_{BR}			V dc
1N6683, 1N6683US					200	
1N6684US, 1N6640, 1N6640US					250	
1N6685, 1N6685US					350	
Reverse current	4016	DC Method	I_{R1}			nA dc
1N6683, 1N6683US		V_R = 175 V dc			100	
1N6684, 1N6684US		V_R = 200 V dc			100	
1N6640, 1N6640US		V_R = 275 V dc			200	
Subgroup 3						
High temperature operation:		T_A = +150°C,	I_{R2}			µA dc
Reverse current	4016	DC method				
1N6683, 1N6683US		V_R = 175 V dc			100	
1N6640, 1N6640US		V_R = 200 V dc			100	
1N6685, 1N6685US		V_R = 275 V dc			200	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Limits			
	Method	Conditions	Symbol	Min	Max	Unit
Subgroup 4						
Capacitance	4001	V_R = 0 V dc V_{sig} = 50 mV(p-p) f = 1 MHz				
1N6683, 1N6683US					5.0	pF
1N6684, 1N6684US					8.0	
1N6685, 1N6685US					8.0	
Reverse recovery time	4031	Condition B $I_F = I_R$ = 30 mA dc I_{RR} = 3.0 mA dc	t_{rr}			nS
1N6683, 1N6683US					50	
1N6684, 1N6684US					50	
1N6685, 1N6685US					50	
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_{FSM} = 3.0 A; 10 surges at 1 per minute (max). Surges duration = 1 μsec				
Electrical measurements		See table I, subgroup 2				

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Thermal shock (glass strain)	1056	Condition A, n = 22, c = 0
Hermetic seal	1071	
Gross leak		
Electrical measurements		See table I, group A, subgroup 2
Subgroup 2		
Not applicable		
Subgroup 3		
Not applicable		
Subgroup 4		
Thermal resistance, junction to lead	3101 or 4081	$L = 3/8"$, $R_{\theta JL} = 160°C/W$ max (see 4.5.3) 1/
Subgroup 5		
Potted environment test	1054	

1/ For US suffix types thermal resistance, junction to end caps. $R_{\theta JEC} = 100°C/W$ max.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

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6.3 Suppliers of JANC die. The qualified JANCH and JANCK suppliers with the applicable letter version will be identified on the QPL (e.g. JANCKA1N6683).

JANC ordering information		
PIN	Manufacturer	
	1N4552	
1N6683 through 1N6685	HA and KA 1N6683 through HA and KA 1N6685	

6.4 Substitution information. The 1N6683 (MIL-S-19500/625) is the preferred type in lieu of the 1N4938 and 1N4938-1 (MIL-S-19500/169) for high reliability, high stress application (see 1.1). The 1N6683 should be used whenever available.

6.4.1 Cross reference substitution list. JANS level will no longer be built to MIL-S-19500/169. Devices in stock are acceptable provided the date code does not exceed the date of implementation of MIL-S-19500/169. Devices required for spec flight applications are found herein. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

Non-preferred PIN	Preferred PIN
1N4938-1	1N6683

6.4.2 Part number marking. JANS devices processed as 1N6683 to the requirments of this specification may alternatively be marked as 1N4938 for a period of three years after the date of this specification. This allowance is for reprocurement only; 1N6683 remains the preferred part number.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - ES

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/625	2. DOCUMENT DATE (YYMMDD) 94/07/15

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING, TYPES 1N6683, 1N6684, 1N6685, 1N6683US, 1N6684US, AND 1N6685US, JANTX, JANTXV, JANS, JANCH, AND JANCK

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial: 513-296-6048 (2) AUTOVON: 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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